

Your grade: 90%

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Next item →

1. Which of the following are true? (Check all that apply.)

1 / 1 point

- ☐ $a_4^{[2]}$ is the activation output of the 2^{nd} layer for the 4^{th} training example
- ☐ X is a matrix in which each row is one training example.
- ☒ X is a matrix in which each column is one training example.

✓ Nice work

- ☐ $a^{[2](12)}$ denotes activation vector of the 12^{th} layer on the 2^{nd} training example.
- ☒ $a_4^{[2]}$ is the activation output by the 4^{th} neuron of the 2^{nd} layer

✓ Nice work

- ☒ $a^{[2]}$ denotes the activation vector of the 2^{nd} layer.

✓ Nice work

- ☒ $a^{[2](12)}$ denotes the activation vector of the 2^{nd} layer for the 12^{th} training example.

✓ Nice work

2. The sigmoid function is only mentioned as an activation function for historical reasons. The tanh is always preferred without exceptions in all the layers of a Neural Network. True/False?

1 / 1 point

- ☒ False
- ☐ True

2. The sigmoid function is only mentioned as an activation function for historical reasons. The tanh is always preferred without exceptions in all the layers of a Neural Network. True/False?

1 / 1 point

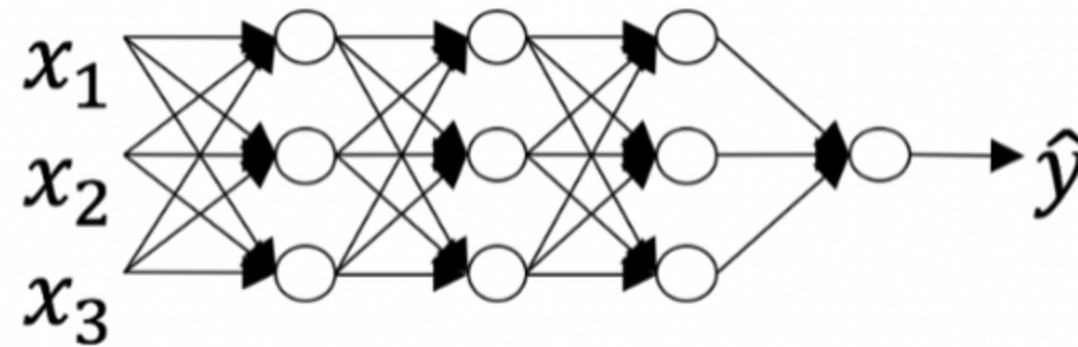
- ☒ False
☐ True

✓ Nice work

Yes. Although the tanh almost always works better than the sigmoid function when used in hidden layers, thus is always proffered as activation function, the exception is for the output layer in classification problems.

3. Which of the following represents the activation output of the second neuron of the third layer applied to the fourth example?

1 / 1 point



- ☒ $a_2^{[3](4)}$
☐ $a_2^{[4](3)}$
☐ $a_4^{[3](2)}$
☐ $a_3^{[4]2}$

✓ Nice work

Yes. The superscript in brackets indicates the layer number, the superscript in parenthesis represents the number of examples, and the subscript the number of the neuron.

4. You are building a binary classifier for recognizing cucumbers ($y=1$) vs. watermelons ($y=0$). Which one of these activation functions would you recommend using for the output layer?

1 / 1 point

- ☐ tanh
- ☒ sigmoid
- ☐ Leaky ReLU
- ☐ ReLU

 **Nice work**

Yes. Sigmoid outputs a value between 0 and 1 which makes it a very good choice for binary classification. You can classify as 0 if the output is less than 0.5 and classify as 1 if the output is more than 0.5. It can be done with tanh as well but it is less convenient as the output is between -1 and 1.

5. Consider the following code:

1 / 1 point

```
A = np.random.randn(4,3)
```

```
B = np.sum(A, axis = 1, keepdims = True)
```

What will be B.shape? (If you're not sure, feel free to run this in python to find out).

- ☐ (4,)
- ☐ (3,)
- ☐ (1, 3)
- ☒ (4, 1)

 **Nice work**

Yes, we use (`keepdims = True`) to make sure that A.shape is (4,1) and not (4,). It makes our code more robust.

6. Suppose you have built a neural network. You decide to initialize the weights and biases to be zero. Which of the following statements is true?

0 / 1 point

6. Suppose you have built a neural network. You decide to initialize the weights and biases to be zero. Which of the following statements is true?

0 / 1 point

- ☐ Each neuron in the first hidden layer will compute the same thing, but neurons in different layers will compute different things, thus we have accomplished “symmetry breaking” as described in the lecture.
- ☐ Each neuron in the first hidden layer will perform the same computation. So even after multiple iterations of gradient descent, each neuron in the layer will be computing the same thing as other neurons.
- ☒ The first hidden layer’s neurons will perform different computations from each other even in the first iteration; their parameters will thus keep evolving in their own way.
- ☐ Each neuron in the first hidden layer will perform the same computation in the first iteration. But after one iteration of gradient descent they will learn to compute different things because we have “broken symmetry”.

[⊗ Try again](#)

7. Logistic regression’s weights w should be initialized randomly rather than to all zeros, because if you initialize to all zeros, then logistic regression will fail to learn a useful decision boundary because it will fail to “break symmetry”, True/False?

1 / 1 point

- ☐ True
- ☒ False

[✔ Nice work](#)

Yes, Logistic Regression doesn’t have a hidden layer. If you initialize the weights to zeros, the first example x fed into the logistic regression will output zero but the derivatives of the Logistic Regression depend on the input x (because there’s no hidden layer) which is not zero. So at the second iteration, the weights’ values follow x ’s distribution and are different from each other if x is not a constant vector.

8. Which of the following is true about the tanh function?

1 / 1 point

- ☒ For large values the slope is close to zero.
- ☐ For large values the slope is larger.
- ☐ The derivative at $c = 0$ is not well defined.
- ☐ The slope is zero for negative values.

8. Which of the following is true about the tanh function?

1 / 1 point

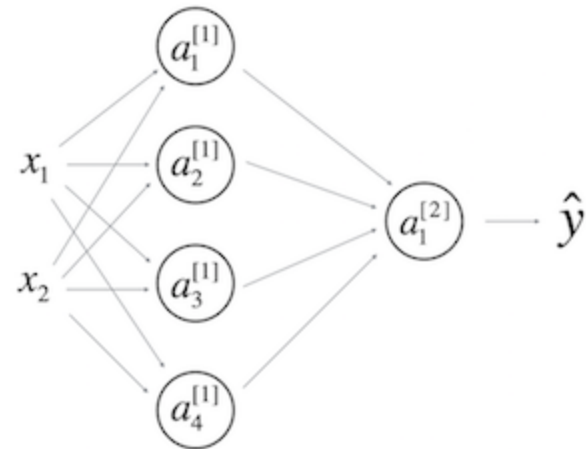
- ☒ For large values the slope is close to zero.
- ☐ For large values the slope is larger.
- ☐ The derivative at $c = 0$ is not well defined.
- ☐ The slope is zero for negative values.

✓ Nice work

Yes. We can see in the graph of the $y = \tanh(c)$ how as the values of c increase the curve becomes flatter.

9. Consider the following 1 hidden layer neural network:

1 / 1 point



Which of the following statements are True? (Check all that apply).

- ☐ $W^{[2]}$ will have shape (4, 1)
- ☐ $b^{[2]}$ will have shape (4, 1)
- ☒ $W^{[2]}$ will have shape (1, 4)

✓ Nice work

9. Consider the following 1 hidden layer neural network:

1 / 1 point



Which of the following statements are True? (Check all that apply).

☐ $W^{[2]}$ will have shape (4, 1)

☐ $b^{[2]}$ will have shape (4, 1)

☒ $W^{[2]}$ will have shape (1, 4)

✔ Nice work

☐ $b^{[1]}$ will have shape (2, 1)

☒ $b^{[1]}$ will have shape (4, 1)

✔ Nice work

☐ $W^{[1]}$ will have shape (2, 4)

☒ $W^{[1]}$ will have shape (4, 2)

✔ Nice work

☒ $b^{[2]}$ will have shape (1, 1)

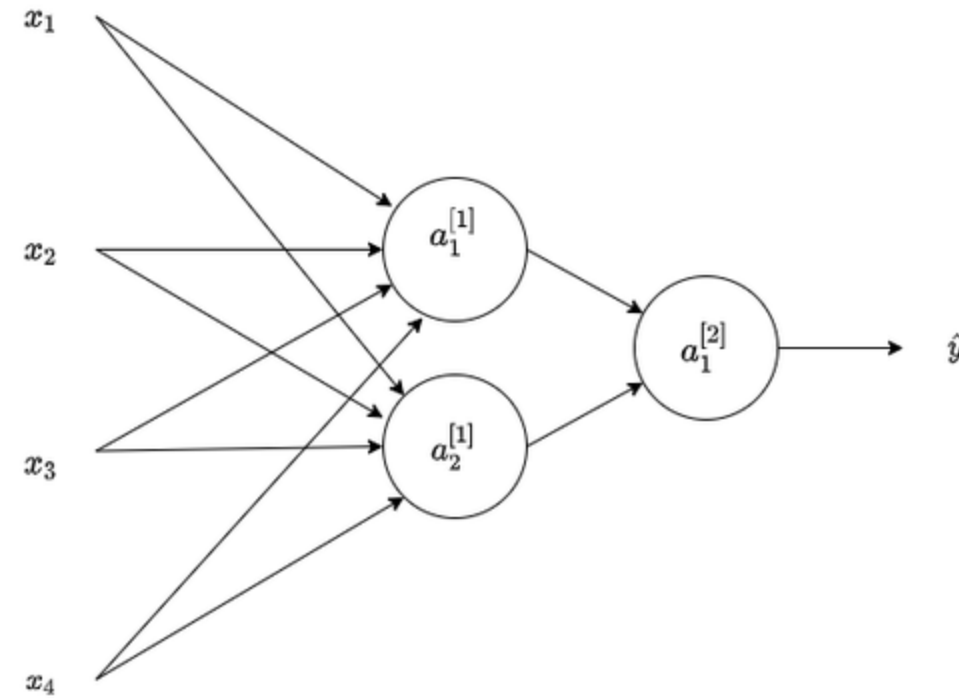
✔ Nice work

10. Consider the following 1 hidden layer neural network:

1 / 1 point

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1 / 1 point



What are the dimensions of $Z^{[1]}$ and $A^{[1]}$?

- ☐ $Z^{[1]}$ and $A^{[1]}$ are (4, 1)
- ☒ $Z^{[1]}$ and $A^{[1]}$ are (2, m)
- ☐ $Z^{[1]}$ and $A^{[1]}$ are (2, 1)
- ☐ $Z^{[1]}$ and $A^{[1]}$ are (4, m)

✔ Nice work

Yes. The $Z^{[1]}$ and $A^{[1]}$ are calculated over a batch of training examples. The number of columns in $Z^{[1]}$ and $A^{[1]}$ is equal to the number of examples in the batch, m. And the number of rows in $Z^{[1]}$ and $A^{[1]}$ is equal to the number of neurons in the first layer.